
A Patient World Model for Early Forecasting of Digital Health Campaign Outcomes: Capabilities and Limits

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Abstract

Digital direct-to-consumer (DTC) health campaigns are usually measured after the fact. In-flight forecasting commonly relies on a separate classifier for every cutoff and horizon. We treat this task as a dynamic-system problem and build a compact patient world model. The architecture maintains a latent state per patient, learns exposure-conditioned state dynamics jointly with a weekly conversion hazard, and rolls forward into future conversion curves. We evaluate it on a US campaign dataset with 147,173 patients and 5.2 million at-risk person-weeks. In a retrospective evaluation conditioned on recorded future exposures, the model forecasts the remaining new-to-brand prescription volume through week 52 with a relative error of 2.9% from a week-4 cutoff and 0.8–2.6% from cutoffs at weeks 8–26. The strongest non-recurrent baseline, a pooled-hazard gradient boosting model given the same survival rollout and information, has relative errors of 13.6–33.1%. Per-horizon classifiers perform substantially worse. A Fisher-information analysis motivates dense next-exposure supervision when conversions are rare. Removing this auxiliary objective increases prescription-volume error by approximately 2–14 $\times$, while providing no consistent disadvantage on the more common specialist-visit outcome. We also evaluate scenario simulation. Switching all future exposure off raises predicted conversion from 0.31 to 0.89, a pattern consistent with selection effects in observational exposure data. This result highlights the limits of interpreting exposure-conditioned rollouts causally.

1 Introduction

Digital direct-to-consumer (DTC) campaigns for prescription drugs produce patient-level exposure records. A patient may see campaign ads repeatedly across channels and ad formats over many weeks. The outcome of interest, such as a specialist visit or a new prescription, may occur months later. Measurement today is mostly retrospective. Attribution and media-mix models [Shao and Li, 2011, Zhang et al., 2014, Jin et al., 2017] are fit after outcomes have matured. Campaign teams cannot wait that long. Budgets, channel mix, and creative rotation must be adjusted mid-flight, using partial data. The common in-flight practice is to select a cutoff week τ and a horizon H , train a classifier for that pair, and repeat for every pair the business needs. This practice has two problems. First, nothing enforces consistency across models trained for different horizons, so the assembled cumulative conversion curve can decrease, which is impossible for a true cumulative curve. Second, none of these classifiers can predict what would happen under a different future exposure plan. They contain no exposure-conditioned dynamics to simulate. World models offer the missing structure. A world model jointly learns how a latent state evolves under actions and predicts the associated outcomes, then rolls forward to simulate future trajectories [Sutton, 1991, Ha and Schmidhuber, 2018]. This framework now spans control [Hafner et al., 2020, 2025], interactive environments [Bruce et al., 2024], video simulation [Brooks et al., 2024, Assran et al., 2025], and driving [Hu et al.,

2023]. Recent position papers identify world models as the next frontier of machine intelligence [LeCun, 2022, Li, 2025]. Patient world models have also begun to emerge, including Delphi-2M [Shmatko et al., 2025] and ETHOS [Renc et al., 2024]. These models generate plausible future health trajectories conditioned on patient history. What they lack is an explicit representation of the intervention. Without one, a trajectory generator cannot answer the question a decision maker actually asks: what happens under this plan? Our setting supplies the missing piece. Exposures are recorded at patient-week granularity, and outcomes are dated events, allowing forecasts to be evaluated against held-out observations at scale.

This paper develops and evaluates an intervention-conditioned patient world model. Our primary endpoint is new-to-brand prescriptions (NBRx), a key business outcome. This outcome is rare, making early forecasting particularly valuable and difficult. In a retrospective evaluation conditioned on recorded future exposures, the model forecasts the remaining NBRx volume through week 52 with a relative error of 2.9% from a week-4 cutoff. Common practice produces errors exceeding 2,500% on the same task. Even a strong pooled-hazard gradient boosting baseline with the same survival rollout has relative errors of 13.6–33.1% across cutoffs at weeks 4–26, compared with 0.8–2.9% for our model. Exposure is targeted rather than randomized, so every forecast in this paper is exposure-conditioned rather than causal. One of our experiments demonstrates the severe prediction reversal that can arise when these scenario rollouts are interpreted causally.

Our contributions are as follows.

1. **An architecture.** An intervention-conditioned patient world model comprising a swappable sequence-state module (here a GRU), a survival hazard head, and an auxiliary next-exposure prediction head. The components are trained jointly, and recursive survival rollout produces coherent conversion curves (Section 3).
2. **An analysis of where the gains arise.** We establish coherence and an additive bound on cumulative-incidence error in terms of weekly hazard errors, derive a concentration bound for aggregate volume under explicit assumptions, and use a Fisher-information argument to explain the potential value of dense auxiliary supervision in the rare-event regime (Section 4).
3. **Accurate early-cutoff forecasts of a rare outcome.** NBRx volume forecasts have relative errors of 0.8–2.9% across the evaluated cutoffs at weeks 4–26, compared with 13.6–33.1% for the pooled-hazard gradient boosting baseline. Removing the auxiliary objective increases volume error by approximately 2–14 $\times$ on this outcome, with no consistent loss on the more common specialist-visit outcome, supporting the predicted regime dependence (Section 6).
4. **A quantified failure of scenario simulation.** Turning all future exposure off raises predicted conversion from 0.31 to 0.89 in our model, with a similar reversal in the baseline simulator. We examine this artifact in relation to exposure selection and show why these observational rollouts cannot be read as causal responses (Section 6.4).

2 Related work

World models. Learning environment dynamics together with the outcome signal, then rolling the model forward, is the core pattern of model-based reinforcement learning [Sutton, 1991, Ha and Schmidhuber, 2018, Hafner et al., 2020, 2025]. The same recipe has scaled to generative interactive environments [Bruce et al., 2024], video models used as general-purpose simulators [Brooks et al., 2024, Assran et al., 2025], and driving [Hu et al., 2023]. We borrow the structure: latent state, action-conditioned transition, joint training, and rollout. We do not borrow the control objective. No policy is optimized, and our actions are observed exposures. The structure buys us measurement: coherent forecasts at every horizon, plus scenario simulation under stated assumptions.

World models for health. Generative models of patient trajectories are emerging fast. Delphi-2M models the natural history of over 1,000 diseases from biobank-scale records [Shmatko et al., 2025]. ETHOS predicts tokenized patient timelines zero-shot [Renc et al., 2024]. MOTOR pre-trains a time-to-event foundation model on structured medical records [Steinberg et al., 2024], and digital-twin models generate synthetic control arms for trials [Fisher et al., 2019]. Surveys note that most EHR foundation models are still evaluated far from decisions [Wornow et al., 2023]. All of these model the outcome process. The intervention process is absent or folded implicitly into the history. Our model is far smaller, but its action stream is explicit and weekly. That is what makes

intervention-conditioned rollout and scenario simulation well-posed tasks. We view the two lines as complementary. Trajectory foundation models supply rich patient states. An explicit intervention-conditioned transition turns a trajectory generator into a decision tool.

Treatment-response forecasting over time. Recurrent marginal structural networks [Lim et al., 2018] and counterfactual recurrent networks [Bica et al., 2020] estimate outcomes under hypothetical treatment sequences. They inherit the assumptions of marginal structural models, such as sequential ignorability [Robins et al., 2000]. We claim less. Our model produces exposure-conditioned forecasts of observed outcomes, and scenario rollouts are simulations under stated assumptions. Section 6.4 shows concretely what goes wrong when this distinction is ignored.

3 An intervention-conditioned patient world model

3.1 Setup and tasks

Consider a campaign observed week by week. For patient i , let $r \in \{0, 1, \dots, R\}$ index the week relative to enrollment, with $R = 52$ in our data. During week r the patient receives digital exposure summarized by an *action* vector $\mathbf{a}_{i,r} \in \mathbb{R}^{d_a}$. It collects log-transformed impression counts by channel, ad type, and targeting tactic, plus recency information. A static vector $\mathbf{x}_i \in \mathbb{R}^{d_s}$ holds eligibility flags, source of business, and baseline activity. The word *action* means recorded observational exposure. Delivery is targeted, not randomized. Every quantity in this paper is therefore an exposure-conditioned forecast, not a causal effect. Section 6.4 makes this concrete.

Let T_i be the week of the conversion event and let C_i be the censoring week, set by the end of the observation window. Following the discrete-time survival convention [Cox, 1972, Tutz and Schmid, 2016], define the per-week hazard

$$h_{i,r+1} = \mathbb{P}(T_i = r+1 \mid T_i > r, \mathbf{x}_i, \mathbf{a}_{i,1:r}), \quad (1)$$

the probability that a patient at risk through week r converts in week $r+1$. A patient contributes one at-risk observation for every week $r < \min(T_i, C_i)$. The cumulative incidence through week k is

$$F_i(k) = 1 - \prod_{m=1}^k (1 - h_{i,m}), \quad (2)$$

which never decreases in k as long as the hazards are valid probabilities. Neural survival models adopt the same convention [Katzman et al., 2018, Lee et al., 2018, Giunchiglia et al., 2018], but they treat the covariate process as given. That choice supports risk scoring and rules out action-conditioned simulation.

Three tasks are studied. **(T1) One-step hazard prediction.** Given the history through week r , predict $h_{i,r+1}$. This is a foundation check. A model that cannot learn the one-step signal has nothing to roll out. **(T2) Early trajectory forecasting.** Given only the history through a cutoff $\tau < R$, forecast the future curve $F_i(k)$ for every patient still at risk, then aggregate to the campaign volume forecast

$$\hat{N}(k) = \sum_{i \in \mathcal{R}_\tau} w_i \frac{F_i(k) - F_i(\tau)}{1 - F_i(\tau)}, \quad (3)$$

Here $\mathcal{R}_\tau$ is the at-risk set at τ . The weights w_i are inverse-sampling weights [Horvitz and Thompson, 1952] that map the modeling cohort back to natural prevalence. **(T3) Scenario simulation.** From cutoff τ , replace the future exposures with a hypothetical plan $\mathbf{a}'_{i,\tau+1:R}$ and recompute (2). The stated assumption is that the learned dynamics hold on the support of the observed data. Task (T2) is where per-horizon practice breaks down, since its per-cell probabilities share no state and need not be monotone. Task (T3) additionally requires an explicit exposure-conditioned transition model, which neither per-horizon classifiers nor outcome-only sequence models have.

3.2 Architecture

The architecture is defined by three components and one inference pattern (Fig. 1): a sequence-state module f_θ , a hazard head $g_\mathcal{R}$, a transition head $g_\mathcal{T}$, and a recursive survival rollout. The sequence module is deliberately generic. We instantiate it as a two-layer GRU [Cho et al., 2014] with hidden

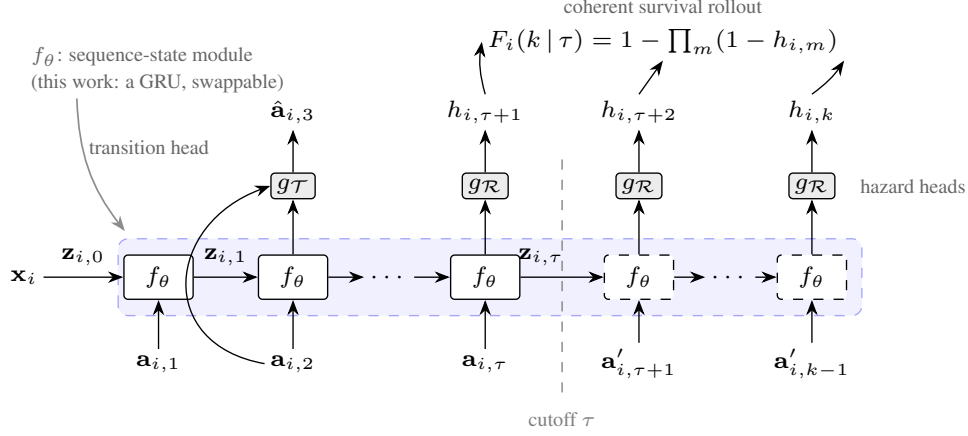


Figure 1: The proposed architecture, unrolled. It is defined by three components and one inference pattern: a swappable sequence-state module f_θ (instantiated as a GRU in this work), a hazard head g_R , a transition head g_T for dense next-action supervision, and the recursive survival rollout that multiplies weekly hazards into a coherent cumulative-incidence forecast. The transition head takes $(\mathbf{z}_{i,r}, \mathbf{a}_{i,r})$, shown by the curved input. During training both heads apply at every at-risk week. The figure shows g_T once on the observed prefix and g_R from the cutoff onward, where forecasting happens. After the cutoff (dashed cells), the same module runs on recorded or hypothetical exposures $\mathbf{a}'$.

size 128 and about 2×10^5 parameters, which trains on CPU inside the certified environments where patient data must stay (Appendix C). Any sequence model can take its place, and a stronger module may improve the results. The GRU is not the contribution. In fact, a plain GRU forecaster is one of our baselines. It is obtained by deleting one component, and Section 6 measures exactly what that deletion costs.

The static vector sets the initial state through a learned projection, and the module advances the state with each week’s action,

$$\mathbf{z}_{i,0} = \tanh(W_0 \mathbf{x}_i + b_0), \quad \mathbf{z}_{i,r} = f_\theta(\mathbf{z}_{i,r-1}, \mathbf{a}_{i,r}), \quad (4)$$

so that $\mathbf{z}_{i,r}$ summarizes what the campaign has done to patient i in the first r weeks. The hazard head is a small two-layer network that outputs a hazard at every at-risk week,

$$h_{i,r+1} = \sigma(g_R(\mathbf{z}_{i,r})). \quad (5)$$

The transition head predicts the next week’s action summary from the current state and action,

$$\hat{\mathbf{a}}_{i,r+1} = g_T(\mathbf{z}_{i,r}, \mathbf{a}_{i,r}). \quad (6)$$

The target of (6) is the observable exposure summary itself. Transition quality can therefore be checked directly on held-out data rather than taken on faith.

3.3 Training objective

Let $\mathcal{A}$ be the set of at-risk patient-weeks (i, r) in the training split, and let $y_{i,r+1} \in \{0, 1\}$ indicate a conversion in week $r+1$. The model is trained end-to-end by minimizing

$$\mathcal{L} = - \sum_{(i,r) \in \mathcal{A}} \left[y_{i,r+1} \log h_{i,r+1} + (1-y_{i,r+1}) \log(1-h_{i,r+1}) \right] + \lambda \sum_{(i,r) \in \mathcal{A}'} \|\hat{\mathbf{a}}_{i,r+1} - \mathbf{a}_{i,r+1}\|_2^2, \quad (7)$$

where $\mathcal{A}'$ keeps the patient-weeks whose next week is observed, and $\lambda = 0.3$ in all experiments. The first term is the discrete-time survival likelihood over pooled at-risk person-weeks [Tutz and Schmid, 2016]. The second term is the dynamics supervision. Setting $\lambda = 0$ and dropping (6) recovers exactly a GRU forecaster of the same capacity. The comparison between the two in Section 6 is therefore a one-flag ablation, not a reimplementaion.

3.4 Rollout and forecasting

Given a cutoff τ , the module reads the observed prefix and produces $\mathbf{z}_{i,\tau}$. The model is then unrolled. At each future week $m > \tau$, the recurrence (4) is advanced with the future action $\mathbf{a}_{i,m}$ and the hazard head outputs $h_{i,m}$. The forecast cumulative incidence from the cutoff is

$$F_i(k | \tau) = 1 - \prod_{m=\tau+1}^k (1 - h_{i,m}), \quad k \in (\tau, R], \quad (8)$$

monotone and consistent across every horizon by construction. The future actions can be the recorded exposures, a stated continuation rule, or a hypothetical scenario $\mathbf{a}'$. Campaign-level forecasts follow by the aggregation in (3). Each task exercises one component. Task (T1) uses the state and the hazard head. Task (T2) adds the rollout. Task (T3) adds the exposure-conditioned transition.

4 Analysis: what the structure predicts

This section analyzes the formulation, not the sequence module. Every statement below holds for any encoder f_θ . Proposition 1 is a property of the survival rollout. Proposition 2 is a property of the aggregation. The Fisher argument is a property of the joint objective (7). Proofs are in Appendix F. Throughout, $\{h_{i,m}\}$ are the true hazards and $\{\hat{h}_{i,m}\}$ are the model's estimates.

4.1 Error propagation through the rollout

One may worry that rollout errors compound multiplicatively with the horizon. The product form (8) rules this out.

Proposition 1 (Coherence and error propagation) *Fix any estimates $\hat{h}_{i,m} \in [0, 1]$ and let $\hat{F}_i(k|\tau)$ be computed by (8). Then (i) $\hat{F}_i(k|\tau)$ is non-decreasing in k and $[0, 1]$ -valued for every realization, and (ii)*

$$|\hat{F}_i(k|\tau) - F_i(k|\tau)| \leq \sum_{m=\tau+1}^k |\hat{h}_{i,m} - h_{i,m}|, \quad \forall k \in (\tau, R]. \quad (9)$$

The trajectory error grows at most additively with the horizon, in increments that are differences of small numbers. Weekly hazards average $\bar{h} \approx 9 \times 10^{-3}$ in our data. Part (i) costs nothing. It holds for any hazard estimates and for any model given the pooled-hazard-plus-rollout form, boosted trees included. The per-horizon practice has neither property. Each (τ, H) classifier estimates $F_i(k)$ on its own, so monotonicity can fail. A plain classifier also has no way to use censored patients. Excluding them selects converters into the long-horizon training slices, a bias that Section 6.1 measures at full size.

4.2 Concentration of the campaign-volume forecast

The next question is when the aggregate forecast (3) can be trusted. Condition on the cutoff-time information $\mathcal{F}_\tau$. For each patient $i \in \mathcal{R}_\tau$, let

$$q_i = \frac{\hat{F}_i(R|\tau) - \hat{F}_i(\tau)}{1 - \hat{F}_i(\tau)}$$

be the predicted probability of converting in the remaining weeks, and let $N = \sum_{i \in \mathcal{R}_\tau} w_i \mathbf{1}\{\tau < T_i \leq R\}$ be the realized weighted converter count.

Proposition 2 (Volume concentration) *Assume (i) conversions of distinct patients are independent conditional on $\mathcal{F}_\tau$, and (ii) the model is trajectory-calibrated on the risk set, i.e., $\mathbb{P}(\tau < T_i \leq R | \mathcal{F}_\tau) = q_i$. Then $\mathbb{E}[N | \mathcal{F}_\tau] = \hat{N}(R)$, and for all $t > 0$,*

$$\mathbb{P}\left(|N - \hat{N}(R)| \geq t \mid \mathcal{F}_\tau\right) \leq 2 \exp\left(-\frac{t^2/2}{V + \frac{1}{3}w_{\max} t}\right), \quad (10)$$

where $V = \sum_{i \in \mathcal{R}_\tau} w_i^2 q_i(1 - q_i)$ and $w_{\max} = \max_i w_i$.

Proposition 2 splits the volume error into a random part and a systematic part. The random part is bounded by (10). The systematic part is the calibration bias that breaks assumption (ii). Three consequences are used in the experiments. Single-digit-percent volume errors are calibration bias, not sampling noise. Per-cell errors of hundreds of percent cannot be noise at this sample size, so they are structural. And once two coherent models are both well calibrated, the campaign-level aggregate stops separating them, even when patient-level discrimination still does. The 1σ relative floor is about $1/\sqrt{n_+}$, with n_+ the expected converter count. Every volume forecast below is read against this floor.

4.3 Dense supervision in the rare-event regime

Finally, consider what the transition loss contributes to the shared encoder. Fix any encoder parameter u and look at the expected curvature of the two loss terms at an at-risk week. For the hazard term, with logit $\eta_{i,r} = g_{\mathcal{R}}(\mathbf{z}_{i,r})$, the Fisher information contribution is

$$\mathcal{I}_{i,r}^{\mathcal{R}}(u) = h_{i,r+1}(1 - h_{i,r+1}) \left(\frac{\partial \eta_{i,r}}{\partial u} \right)^2 \leq \bar{h} \sup_{i,r} \left(\frac{\partial \eta_{i,r}}{\partial u} \right)^2. \quad (11)$$

The factor $h(1 - h)$ is small because events are rare. Under a Gaussian working model for the transition residuals with variance σ^2 , the transition term contributes

$$\mathcal{I}_{i,r}^{\mathcal{T}}(u) = \frac{\lambda}{\sigma^2} \left\| \frac{\partial \hat{\mathbf{a}}_{i,r+1}}{\partial u} \right\|_2^2, \quad (12)$$

which does not depend on $\bar{h}$ and appears at every observed week, across all d_a components. In our data $\bar{h} \approx 9 \times 10^{-3}$ for the diagnosis outcome and an order of magnitude smaller for the prescription outcome. Per person-week, the outcome term then supplies at most about 1% of the squared-gradient scale available to the encoder, while (12) is $O(1)$. When events are rare, the next-state loss provides most of the curvature that pins down the shared representation.

4.4 Three testable predictions

P1 (structure). Coherence and additive error propagation belong to the rollout, not the network. Any pooled-hazard model with the rollout, including a plain GBM, should beat the per- (τ, H) practice by a wide margin.

P2 (aggregate ceiling). Once coherent models are well calibrated, their volume errors should cluster near the sampling floor. Aggregates then stop separating architectures, while patient-level discrimination still can.

P3 (regime dependence). The value of the learned state and of the dense transition supervision should grow as the outcome gets rarer and as the sequences carry more dynamics. On the common diagnosis outcome the world model should be close to a strong pooled-hazard baseline. On the rare prescription outcome it should pull far ahead.

The headline experiment sits exactly where P3 predicts the largest gap. The supporting results for P1 and P2, where the world model is predicted not to dominate and indeed does not, are in Appendix D.

5 Experimental setup

Data. The data come from a large US digital DTC campaign for a prescription cardiovascular drug. The records are protected health information. They stay on a secured analytics cluster and only aggregate metrics are reported. From a universe of 10.9M eligible patients, a stratified case-control cohort of 147,173 patients is built and split 70/15/15 by patient hash. The pooled at-risk person-week count is 5.2M. The weekly action vector ($d_a = 17$) holds winsorized log impression counts in total and by channel, ad-type, and targeting buckets, an activity flag, and the normalized relative week. The static vector ($d_s = 7$) holds source-of-business indicators, eligibility flags, baseline visit volume, and the enrollment week. Two outcomes are studied. The diagnosis outcome (dx, a visit to a relevant specialist) has about 49k events in the cohort. The prescription outcome (rx, a new-to-brand prescription) has 5,225 events and is roughly ten times rarer. By prediction P3, rx is where the world model should matter most. Two design facts matter when reading the results. The cohort

Table 1: The headline task: forecast the campaign’s final NBRx volume from an early cutoff τ (rx outcome, test split, known future exposures). Relative error (%) of the forecast final converter count. “Coherent” is the fraction of monotone per-patient curves (range across cutoffs). Test risk sets hold 21,963 down to 16,940 patients. Observed future converters $n_+ = 711, 601, 489, 206, 76$. At $\tau=39$ the 1σ sampling floor is $1/\sqrt{76} \approx 11.5\%$, so that column cannot rank the coherent models (see text). All numbers as-run.

Model	NBRx volume rel. error (%) at cutoff τ					Coherent
	$\tau=4$	$\tau=8$	$\tau=13$	$\tau=26$	$\tau=39$	
Per- (τ, H) GBM	2582.0	3054.8	3754.7	8491.6	18389.9	19–24%
Per- (τ, H) GBM + future	2581.8	3054.1	3754.0	8491.9	18389.9	11–31%
Naive extrapolation	64.4	82.6	97.4	166.1	146.7	100%
Pooled-hazard GBM	13.6	17.3	19.2	33.1	44.3	100%
GRU forecaster ($\lambda=0$)	5.2	7.4	8.2	11.1	5.6	100%
GRU world model (proposed)	2.9	2.3	2.6	0.8	23.4	100%

is enriched to 33.6% dx prevalence on purpose, and every metric is reported both on the enriched cohort and reweighted to natural prevalence with the inverse-sampling weights w_i . The sequences are sparse, with a median of three active weeks per patient. Cohort construction and full feature definitions are in Appendices A and B.

Baselines, protocol, and metrics. Every model sees the same features, splits, and test patients. The trajectory baselines are: **per- (τ, H) GBM** and **GBM+future**, LightGBM [Ke et al., 2017] trained separately for each cutoff-horizon cell, reflecting common practice, with the +future variant also given aggregated future exposures. **Pooled-hazard GBM**, a single discrete-time pooled-hazard LightGBM rolled out exactly like our model. It is one coherent model for all horizons, added on purpose so that the proposed model does not win by strawman. **GRU forecaster**, the $\lambda = 0$ ablation. **Naive**, which extrapolates the campaign-to-date constant hazard. The one-step task additionally uses logistic regression, flat LightGBM, and an MLP (Appendix D). Only the proposed model is tuned, on validation hazard likelihood. All baselines run standard library defaults and are reported as they ran. No baseline was weakened. Campaign-level metrics per cutoff are the relative *volume error* of the final converter count (3) against the Kaplan–Meier [Kaplan and Meier, 1958] count on the risk set, the *trajectory calibration* (ICI, the mean absolute gap between predicted and observed curves), and *coherence* (the fraction of monotone per-patient curves). Patient-level metrics are AUROC, AUPRC, and calibration, both enriched and IPW-corrected. The transition head is scored by held-out next-state skill against mean and persistence predictors, so the claim that it learns dynamics is checked rather than asserted.

6 Results

6.1 The headline: calling the year-end prescription volume at week 4

The task: after observing the first τ weeks, forecast the number of new-to-brand prescriptions that the at-risk audience will generate by week 52. The evaluation feeds every model the recorded future exposures, so the comparison isolates learned response dynamics. Forecasts are per-patient curves (8) aggregated by (3) and scored against the Kaplan–Meier count on the same risk set. Table 1 is the exact record. Fig. 2 plots the same numbers against the sampling floor. At $\tau = 4$, with 711 future converters hidden in a risk set of 21,963 test patients, the world model forecasts 691 conversions against 711 observed, an error of 2.9%. From $\tau = 26$ it forecasts 208 against 206 (0.8%). Every curve is monotone by construction.

How close is that to the best possible? By Proposition 2, even a perfectly calibrated forecaster faces a sampling floor of about $1/\sqrt{n_+}$, because the realized converter count is itself random. On these risk sets the 1σ floor is 3.7%, 4.1%, 4.5%, and 7.0% at $\tau = 4$ –26. The world model’s errors of 2.9, 2.3, 2.6, and 0.8% sit at or below the floor at every one of these cutoffs. On this test set it is statistically indistinguishable from a perfectly calibrated forecaster. There is no headroom left to measure. The pooled-hazard GBM sits 3.6–4.7 σ above the floor at the same cutoffs, which is systematic bias rather than noise.

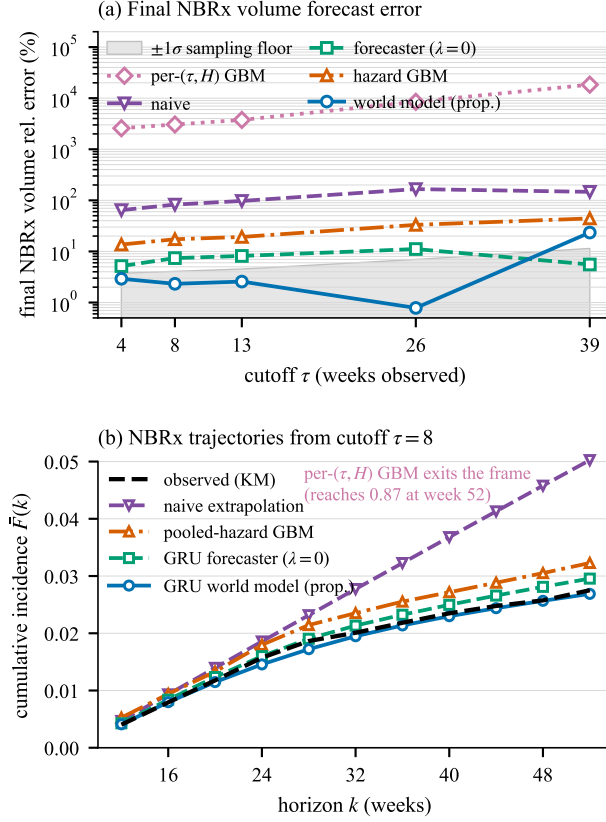


Figure 2: The headline result (rx outcome, test split). (a) Relative error of the forecast final NBRx volume versus the cutoff τ (log scale). The shaded band is the $\pm 1\sigma$ sampling floor implied by Proposition 2 ($\approx 1/\sqrt{n_+}$). No method, however good, can be expected to beat it. The world model sits inside the floor at every cutoff up to week 26. The fair coherent baseline sits 4–5 σ above it, and the per-horizon practice two to four orders of magnitude above it. Read horizontally, a small τ means a long rollout ($\tau = 4$ rolls 48 weeks), and the world model’s error does not grow with rollout length, as Proposition 1 requires. (b) Predicted NBRx cumulative-incidence trajectories from cutoff $\tau = 8$ against the observed curve, estimated by Kaplan–Meier (KM), the standard estimator of cumulative incidence under right-censoring. The world model tracks the observed curve across all 44 remaining weeks. The whole curve is calibrated, not just the endpoint. The per-cell curve leaves the frame.

No baseline produces a usable number for this task. The per-horizon practice errs by 2,582% at week 4 and by 18,390% at week 39, forecasting about 19,000 conversions where 711 occur. The mechanism is censoring. A per-cell classifier can label a non-converter at horizon k only if that patient is observed through week k . Its long-horizon training slices therefore keep every converter but only the earliest-enrolled non-converters. The slices become converter-dominated, the classifier learns a conversion rate near 0.9, and it applies that rate to a risk set whose true rate is 0.03. The same method errs an order of magnitude less on the ten-times-more-common dx outcome (Appendix D), which is exactly how a selection-through-censoring bias should scale. Fewer than a third of the assembled curves are monotone. Giving these models the aggregated future exposures changes nothing, so the failure is structural rather than informational. The pooled-hazard GBM is the strong baseline built to prevent a strawman win. It shares the survival rollout, the features, and the future exposures. Structure alone carries it to 13.6–33.1%, far beyond common practice, yet an order of magnitude behind the world model at every cutoff. The world model’s trajectory ICI is also three to six times tighter, and both GRU models add about +0.02 AUROC over the GBM baseline (Appendix D). Read vertically, Fig. 2(a) is an ablation ladder. Coupling the horizons at all is worth a factor of 40. The survival rollout is worth another 5. The learned state is worth another 3, and

the transition supervision a further 2–14 $\times$, dissected next. Panel (b) shows that the whole curve is calibrated, so one trained model answers every interim milestone question with the same accuracy.

Two qualifications. First, at $\tau = 39$ only 76 future converters remain and the floor rises to 11.5%. The world model’s 23.4% is about 2σ and the two GRU variants differ by less than 1.5σ , so that column cannot rank the coherent models. A real late-cutoff effect does exist on dx, where both GRU variants underpredict by about 15% while the GBM stays at 6%. It hits both values of λ equally, so it is not about the transition loss. Appendix D analyzes it. Second, this is a retrospective evaluation with known future exposures, applied equally to every model. Section 6.3 discusses the step to a fully operational forecast.

6.2 Why the world model wins here, and only here

The headline number would be suspicious if the world model won everywhere. Strong methods usually have a regime, and the analysis predicted this one in advance. Two checks confirm the mechanism.

First, the gap appears exactly where P3 puts it. On the diagnosis outcome, with ten times more events on the same patients and pipeline, the three coherent models land in the same single-digit band. The world model reaches 4.5–7.5% for $\tau \leq 26$ and the GBM baseline 6.0–9.2%, which is the aggregate ceiling of P2. The per-cell practice still fails by 209–1573%, as P1 predicts. The full dx tables, the one-step task, and a dynamics-rich subgroup analysis are in Appendix D. The decisive advantage is confined to the rare outcome. That is what the Fisher argument requires. With $\bar{h}$ an order of magnitude smaller, the outcome likelihood alone leaves the encoder loosely determined, so the models that share the dense supervision should separate from those that do not.

Second, the one-flag ablation isolates the ingredient. Setting $\lambda = 0$ removes exactly one thing, the transition loss. On dx the ablation costs nothing, and forecaster and world model are statistically inseparable there. On rx it costs 2–14 $\times$ in volume error (5.2/7.4/8.2/11.1% against 2.9/2.3/2.6/0.8% at $\tau = 4$ –26), with discrimination essentially tied. One flag, one order of magnitude, only in the rare-event regime, in the direction predicted by (11)–(12) before the experiment ran. We read this as supportive rather than conclusive, since it is one outcome on one campaign and the noisy $\tau = 39$ cell goes the other way. Still, the location and the size of the effect are hard to explain by anything except the dense supervision.

For completeness, two dead ends are reported so others do not re-run them. A 30k-patient probe showed a 2–5 $\times$ world-model advantage on dx that vanished at full scale, a small-sample effect. A multi-step rollout training objective made calibration worse and was removed (Appendix C).

6.3 From retrospective to operational forecasting

The evaluations above feed the recorded future exposures to every model. This is the right controlled comparison, because it isolates response dynamics from the separate problem of knowing the future media plan. An operational week-4 forecast does not know that plan.

The world model is the only method in Table 1 with a mechanism for closing this gap from within. Its transition head (6) predicts next week’s exposure with +11.5% held-out skill over a mean predictor, so the model can roll forward on its own predicted exposures. Every baseline must instead be handed an assumed future plan, and the assumption is load-bearing. The natural choice of assuming no further exposure is exactly the off-support scenario that Section 6.4 shows to inflate predicted conversion by a factor of two or more. A closed-loop evaluation of this capability is running at the time of writing and will be reported in a revision. The present version claims only the controlled result.

6.4 Scenario simulation and a selection artifact

The transition model also turns the forecaster into a simulator. Replace the future exposures with a stated scenario $\mathbf{a}'$, re-roll, and read off the predicted trajectory. We exercised this with dose scaling ($\mathbf{a}' = \alpha \mathbf{a}$, $\alpha \in [0, 4]$), per-lever shutoffs, and a placebo control, on dx at $\tau \in \{4, 8\}$ (protocol in Appendix E). Within the observed dose support the simulator behaves sanely (Table 2). The dose-response is monotone and saturating but small, about +0.6 percentage points for a 4 $\times$ dose. The placebo moves the forecast by less than 10^{-4} , and no single lever moves it by more than 0.003.

Table 2: Scenario simulation (T3): simulated dose-response on the dx outcome. Each cell is the mean predicted cumulative conversion of the test risk set when all future exposure is scaled by α ($do(\alpha \cdot \mathbf{a})$), for the world-model simulator and the perturbed pooled-hazard GBM. The $\alpha = 0$ column is a selection artifact, not an effect (see text). The $\alpha = 4$ column is outside the observed dose support.

Simulator (τ)	Dose scale α						
	0	0.25	0.5	1	1.5	2	4
World model ($\tau=4$)	0.893	0.312	0.310	0.310	0.311	0.311	0.316
Pooled-hazard GBM ($\tau=4$)	0.632	0.315	0.314	0.313	0.318	0.325	0.364
World model ($\tau=8$)	0.708	0.267	0.265	0.265	0.266	0.267	0.270
Pooled-hazard GBM ($\tau=8$)	0.504	0.273	0.272	0.271	0.275	0.279	0.307

The instructive result is off support. Setting $\alpha = 0$ switches all future exposure off, and on a causal reading this should produce the lowest conversion. It produces the highest by far. Predicted conversion jumps from 0.31 to 0.89. The mechanism is selection through censoring. In the observed data a patient’s exposure stops once they convert, so no future exposure is strong evidence of imminent conversion. The GBM baseline simulator shows the same inversion at 0.63, so this is a property of observational exposure data under outcome-dependent censoring, not a defect of one architecture. The number is worth remembering. It is the size of the gap between what patients who look like this do and what would happen if we did this. Turning scenario outputs into effect estimates requires design, such as holdouts, randomization, or explicit deconfounding [Robins et al., 2000, Künzel et al., 2019].

7 Conclusion

We formulated in-flight campaign measurement as exposure-conditioned patient-state trajectory modeling and built a compact patient world model for it. From four weeks of observation, the model calls the campaign’s final volume on the rare business endpoint within 2.9%, at the sampling floor of the test set. This is a task on which standard practice is off by orders of magnitude. The mechanism is understood rather than guessed. A Fisher-information argument locates the value of dense transition supervision in the rare-event regime, and a one-flag ablation confirms both the location and the size of the effect. The same recipe, a survival-coherent rollout from a learned intervention-conditioned state, applies wherever longitudinal exposures meet sparse terminal outcomes. Adherence programs, vaccination campaigns, and public-health communication are natural next targets.

Responsible use and limitations

This work forecasts observed health-seeking behavior under recorded digital exposure. It does not estimate causal effects, and Section 6.4 quantifies why its scenario outputs must not be read as such. Model outputs are decision support for campaign measurement and planning. They are not used, and should not be used, for individual-level clinical or coverage decisions. Uncertainty is reported throughout against the sampling floor of Proposition 2. Forecasts at cutoffs whose floor exceeds the model differences are presented as unrankable rather than interpreted, and a known late-cutoff underprediction bias of the recurrent models is documented in Appendix D. All results come from one campaign, one therapeutic area, and one country, on an intentionally enriched cohort that is reweighted to natural prevalence at evaluation. External validity is untested. The trajectory evaluations condition on recorded future exposures, applied equally to every model. An operational closed-loop variant is under evaluation (Section 6.3). The underlying data are protected health information analyzed inside a certified environment. Only aggregate metrics are reported, and no patient-level output leaves that environment. Extended limitations and deployment guidance are in Appendices D and G.

Reproducibility and data availability. The underlying data cannot be shared. Cohort construction rules, feature definitions, hyperparameters, and the full as-run result tables are given in Appendices A–G, at a level of detail sufficient to rebuild the pipeline on comparable data.

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A Cohort construction

The modeling universe contains 10.88M patients with campaign eligibility. The stratified case-control cohort keeps: all rx converters (5,225), all on-market comparator patients (4,399), a deterministic-hash subsample of dx converters (44,917), and a deterministic-hash subsample of on-market non-converting negatives with at least eight observed weeks (95,061). The total is 147,173 patients. Deterministic hashing (no random seed) makes the cohort exactly reproducible from the universe. Each patient carries $w_i = 1/(\text{stratum sampling fraction})$, equal to 1 for the kept-all strata. Evaluation-time inverse-probability weighting with w_i restores natural prevalence [Horvitz and Thompson, 1952].

Calendar weeks are indexed as $wk = \lfloor (date - 2024-01-01)/7 \rfloor$. The relative week is $r = wk - wk_{index}$. The observation horizon is $r \in [0, 52]$ with a fixed absolute window end, so each patient’s observed relative window is $\min(52, 53 - wk_{index})$. Late enrollers have shorter windows, handled as right-censoring in the survival likelihood. After a conversion event the patient exits the risk set. Splits are 70/15/15 by patient hash, and no patient appears in more than one split. The dynamics-rich subgroup used in Appendix D is defined as patients with at least five total active exposure weeks (29,859 patients, 20.3%). The originally intended definition of eight or more active weeks is nearly empty under this cohort’s sparsity. It was widened before any model comparison was made on the subgroup.

B Feature definitions

Per-week action vector ($d_a = 17$). $\text{imp_log}(\log(1 + x))$ of the p99-winsorized total weekly impressions, $\log(1 + \cdot)$ of the weekly event count, $\log(1 + \cdot)$ impression volumes in three channel buckets (digital, programmatic, display), three ad-type buckets (display, video, audio), seven targeting buckets (condition, lookalike, behavioral-contextual, conquensing, retargeting, geo-demographic, none/other), a week-active flag, and the normalized relative week $r/52$.

Static vector ($d_s = 7$). Source-of-business one-hot (three levels: market-naive, switched from another product, continuing on brand), rx-eligibility flag, dx-eligibility flag, $\log(1 + \cdot)$ prior office-visit count, and normalized enrollment week.

Survival convention. The hazard emitted from state $\mathbf{z}_{i,r}$ predicts an event in week $r+1$. At-risk person-weeks are pooled for training and evaluation. Patients are censored at their observation end.

C Training details

The proposed model is implemented in PyTorch and trained on CPU. The tuned configuration (selected on validation hazard likelihood) uses two GRU layers, hidden size 128, dropout 0.2. The base configuration is one layer, hidden size 64. The one-step task (Table 3) uses the base configuration. The rollout and simulation experiments use the tuned configuration. The static projection W_0 maps the static vector to the initial hidden state of every layer. The hazard head is hidden $\rightarrow$ hidden $\rightarrow$ 1 with ReLU and dropout. The transition head maps $(\mathbf{z}_{i,r}, \mathbf{a}_{i,r})$ through one hidden layer to the 17-dimensional next-week action summary. The loss weight is $\lambda = 0.3$. Optimization: Adam, learning rate 10^{-3} , ReduceLROnPlateau (factor 0.5, patience 2) on validation hazard negative log-likelihood, early stopping (patience 10), at most 50 epochs, batch size 512. A control run at batch size 1024 produced materially identical results, so batch size is not a sensitive choice here. A multi-step rollout training objective (supervising the cumulative incidence along the rollout rather than only one-step hazards) was tried and made calibration worse. It is not used anywhere in this paper. The GRU forecaster ablation sets $\lambda = 0$ and removes the transition head, with nothing else changed. On the rollout task the transition head reaches +11.5% held-out skill over the training-mean predictor, on both outcomes. Baselines use standard library defaults (scikit-learn for logistic regression and MLP, LightGBM [Ke et al., 2017] for all GBM variants) and are reported as-run.

D Full as-run results on the diagnosis outcome and subgroups

This appendix reports everything the main text compresses: the one-step foundation task, the full dx trajectory tables, and the dynamics-rich subgroup, all as-run.

D.1 One-step hazard prediction (T1)

On the one-step task the world model is best on every metric, but the margins over LightGBM and the MLP are small, as expected. With a median of three active weeks per patient, cumulative features already capture most of the history, and the one-step task does not exercise the rollout. On the rx outcome (748 test positives) all models land at AUROC 0.84–0.86, within noise of one another. The one-step likelihood cannot separate architectures there, consistent with Section 4.3.

Table 3: One-step hazard prediction (T1), dx outcome, full test split (22,086 patients, 768,431 at-risk person-weeks, 6,856 positives). AUROC is reported on the enriched cohort and IPW-corrected to natural prevalence. AUPRC is on the enriched cohort.

Model	AUROC (enr.)	AUROC (IPW)	AUPRC (enr.)	Cal. slope	ECE
Base rate	0.500	0.500	0.009	—	—
Logistic regression	0.855	0.838	0.051	1.005	0.0011
LightGBM (flat)	0.872	0.855	0.059	0.648	0.0006
MLP	0.873	0.856	0.065	0.929	0.0009
GRU world model (proposed)	0.880	0.864	0.067	0.957	0.0002

D.2 Trajectory forecasting on dx (T2)

Table 4: Campaign-level trajectory forecasting (T2), dx outcome, test split, full cohort. Volume relative error (%) at the campaign end and coherence. Test risk sets: 20,396 down to 12,805. $n_+ = 5,874$ down to 631.

Model	Volume rel. error (%) at cutoff τ					Coherent
	$\tau=4$	$\tau=8$	$\tau=13$	$\tau=26$	$\tau=39$	
Per- (τ, H) GBM	209.3	256.0	326.7	645.1	1572.8	54–70%
Per- (τ, H) GBM + future	209.2	255.8	326.6	644.9	1572.9	27–72%
Naive extrapolation	97.3	96.8	101.9	120.3	140.5	100%
Pooled-hazard GBM	8.7	9.2	8.8	6.0	6.2	100%
GRU forecaster ($\lambda=0$)	6.6	5.4	3.4	5.7	15.5	100%
GRU world model (proposed)	7.5	6.5	4.8	4.5	15.3	100%

Table 5: Patient-level discrimination and trajectory calibration on dx (T2), full cohort. AUROC is IPW-corrected. ICI is the mean absolute gap between predicted and KM-observed cumulative-incidence curves.

Model	AUROC (IPW)					Trajectory ICI				
	4	8	13	26	39	4	8	13	26	39
Per- (τ, H) GBM	.718	.675	.695	.664	.544	.139	.160	.201	.326	.533
Naive extrapolation	.500	.500	.500	.500	.500	.149	.126	.114	.083	.050
Pooled-hazard GBM	.912	.908	.903	.896	.891	.012	.012	.010	.005	.004
GRU forecaster ($\lambda=0$)	.935	.930	.924	.917	.904	.010	.008	.006	.002	.004
GRU world model (proposed)	.935	.930	.925	.917	.909	.012	.011	.008	.002	.004

Tables 4–5 and Fig. 3 show the dx outcome in full. The reading is the one predicted in Section 4.4. The structural gap against the per-cell practice is enormous (P1). The three coherent models are within a few percent of each other on volume and ICI, near the $\sim 1\%$ sampling floor of Proposition 2 (P2). The world model’s patient-level edge over the pooled-hazard GBM is a stable $+0.018$ to $+0.023$ AUROC at every cutoff.

The late-cutoff underprediction. At $\tau = 39$ both GRU variants underpredict the dx volume by about 15% while the GBM baseline stays at 6%. With $n_+ = 631$ the 1σ floor is only 4%, so this is a systematic effect of about 4σ . Both values of λ are hit identically, so it is a property of the recurrent architecture, not of the transition loss. Two mechanisms plausibly combine. First, training support thins at late relative weeks. The observation window is $\min(52, 53 - \text{wk}_{\text{index}})$, so weeks $r > 39$ are observed only for the earliest enrollees, and the hazard head sees few at-risk examples there. Second, by week 39 most at-risk patients have been exposure-inactive for many consecutive weeks. Under a long constant input the recurrent state drifts toward a fixed point. Inactivity also correlates with having already converted in the training data, the same selection mechanism in mild form that produces the $\alpha = 0$ artifact of Section 6.4, so the drifted state carries a suppressed hazard. The GBM baseline is immune to both, because its cumulative features simply freeze. The practical reading is that the world model’s advantage is a property of early cutoffs, which is where early forecasting

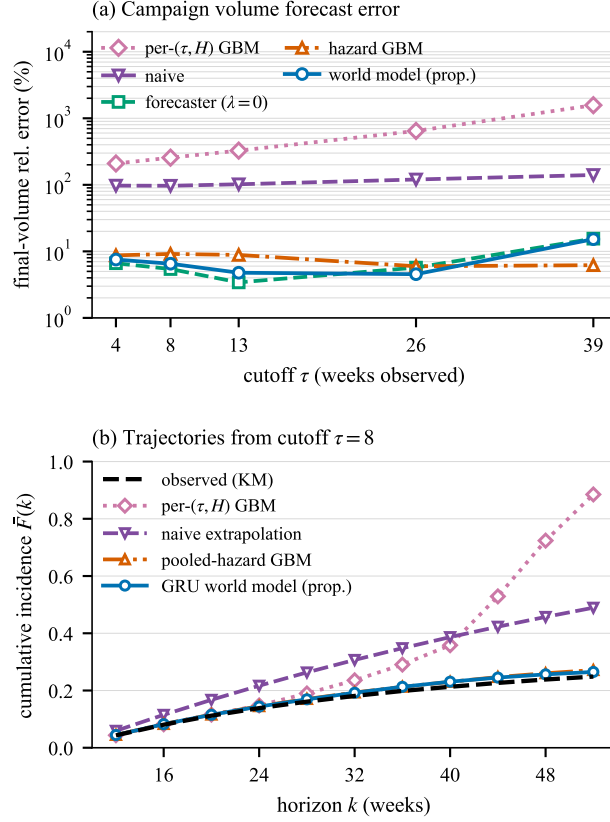


Figure 3: Trajectory forecasting on dx (test split). (a) Final-volume relative error versus cutoff (log scale). (b) Mean predicted cumulative-incidence trajectories from $\tau = 8$ against the Kaplan–Meier curve. The coherent models sit in the same single-digit band. The per-cell practice explodes with the horizon.

lives. A deployment would hand very late cutoffs to the cheaper pooled-hazard model or recalibrate the tail.

D.3 Dynamics-rich subgroup

Table 6: The regime map (volume rel. error, %): the three coherent models on the full dx cohort, the dynamics-rich dx subgroup (≥ 5 total active weeks), and the full rx cohort. Small- n_+ cells are noisy.

Regime	Model	$\tau=4$	$\tau=8$	$\tau=13$	$\tau=26$	$\tau=39$
dx, full cohort (n_+ : 5874–631)	Pooled-hazard GBM	8.7	9.2	8.8	6.0	6.2
	GRU forecaster	6.6	5.4	3.4	5.7	15.5
	GRU world model	7.5	6.5	4.8	4.5	15.3
dx, dynamics-rich (n_+ : 599–91)	Pooled-hazard GBM	15.0	15.2	19.5	21.5	36.1
	GRU forecaster	2.3	0.7	0.3	7.8	16.9
	GRU world model	4.0	1.6	3.2	6.8	17.6
rx, full cohort (n_+ : 711–76)	Pooled-hazard GBM	13.6	17.3	19.2	33.1	44.3
	GRU forecaster	5.2	7.4	8.2	11.1	5.6
	GRU world model	2.9	2.3	2.6	0.8	23.4

On patients with at least five total active weeks (20.3% of the cohort), the pooled-hazard GBM degrades to 15–36% volume error while the GRU models stay at 0.3–18%, a $2\text{--}10\times$ gap, with

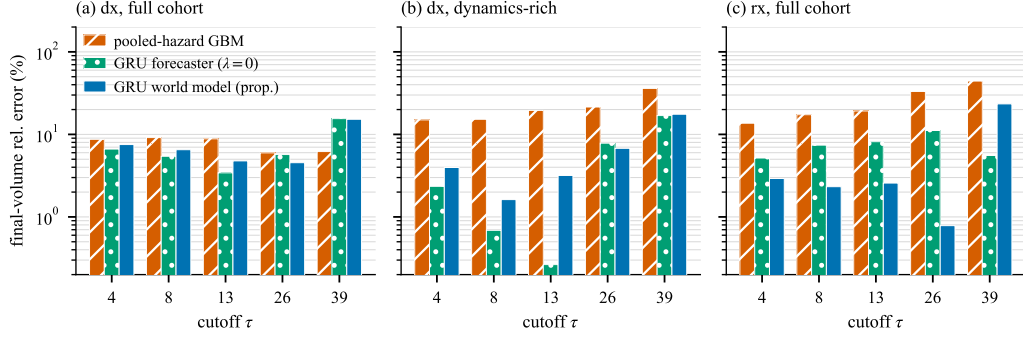


Figure 4: The regime map (Table 6, drawn): final-volume relative error of the three coherent models, log scale. (a) Full sparse dx cohort: the aggregate ceiling of P2. (b) Dynamics-rich patients and (c) the rare rx outcome: the regime dependence of P3.

trajectory ICI $2-3 \times$ lower and about $+0.02$ AUROC on the same patients (Table 6, Fig. 4). Where there is a sequence worth reading, reading it beats summarizing it. Between the two GRU models the subgroup comparison is mixed and within noise, consistent with the main-text finding that transition supervision separates them only in the rare-event regime.

E Scenario-simulation details (T3)

The simulator is the tuned world model (45 epochs, otherwise as in Appendix C), applied to the full test risk sets at $\tau \in \{4, 8\}$ on the dx outcome. The perturbed pooled-hazard GBM serves as the reference simulator. Dose scaling multiplies all future exposure magnitudes by α . At $\alpha = 0$ this also zeroes the week-active flag, which is what triggers the selection artifact. Per-lever shutoffs zero one action dimension’s future values (13 levers: channel, ad-type, and targeting buckets). The largest observed shift was $+0.0026$ (total digital volume off), and the placebo lever (a negligible bucket scaled $4 \times$) moved the forecast by less than 10^{-4} . Per-patient dose-response monotonicity over all α including 0 holds for 20–29% of patients, dragged down by the $\alpha = 0$ artifact. Two protocol refinements are noted for future runs: hold the week-active pattern fixed at $\alpha = 0$ so that only magnitude varies, and restrict reported curves to the observed dose support (approximately $\alpha \in [0.25, 2]$).

F Proofs

Proof of Proposition 1. Part (i) is immediate, because every factor of the product in (8) lies in $[0, 1]$. For part (ii), write the survival functions $\hat{S}(k) = \prod_{m=\tau+1}^k (1 - \hat{h}_{i,m})$ and $S(k) = \prod_{m=\tau+1}^k (1 - h_{i,m})$, and set $P_m^- = \prod_{j=\tau+1}^{m-1} (1 - \hat{h}_{i,j})$ and $P_m^+ = \prod_{j=m+1}^k (1 - h_{i,j})$. Telescoping,

$$\hat{S}(k) - S(k) = \sum_{m=\tau+1}^k P_m^- (h_{i,m} - \hat{h}_{i,m}) P_m^+.$$

Every P_m^- , P_m^+ lies in $[0, 1]$, so the triangle inequality gives (9), and $|\hat{F} - F| = |\hat{S} - S|$. $\square$

Proof of Proposition 2. Unbiasedness follows from assumption (ii) and linearity. The variables $w_i(\mathbf{1}\{\tau < T_i \leq R\} - q_i)$ are conditionally independent by assumption (i), zero-mean, bounded by $w_{\max}$, with variance $w_i^2 q_i(1 - q_i)$. Inequality (10) is Bernstein’s inequality applied to their sum. $\square$

G Practical guidance

The findings compress into three pieces of advice for teams that measure campaigns in flight.

1. For the rare endpoint that matters, the world model is not an increment. It is the difference between having a number and not. A week-4 NBRx volume call within 3% did not previously exist at any price. Common practice is off by orders of magnitude, and even a well-built coherent GBM is off by 14–33%.

2. For common outcomes, fix the formulation and stop. Most of the gap between practice and the world model on the diagnosis outcome is closed by the survival rollout alone, which any pooled-hazard model can wear, including the GBM a team already has. The Fisher calculation of Section 4.3 tells a team in advance which of their outcomes justifies the sequence model.

3. Simulate for planning, not for effect estimation. The simulator answers what the model expects under a stated plan, coherently and per patient. On targeting-driven data its off-support scenarios measure selection rather than response, and the 0.31-to-0.89 inversion is the concrete warning. Scenario outputs need support flags and negative controls. Effect claims need experimental design.